Source and Modifications Disclosure

Smart Home Automation Project

Prepared as a submission-ready account of third-party inputs, project-specific changes, and remaining attribution checks. The available project records identify four Tinkercad feature circuits, a separate water-level monitor board, five circuit reference PDFs, two Arduino sketches, and a Fusion 360 room-and-tank model. They document substantial integration and functional changes, but do not establish the authors, license terms, exact versions, or provenance of every file. Those items are explicitly marked for team confirmation.

Executive disclosure

The project records show that the team's smart-home work drew on four pre-existing Tinkercad circuit designs for temperature-based fan control, light sensing, automatic entry, and PIR motion detection, and on a supplied water-level monitoring board as the starting point for the tank subsystem [R1–R5]. These are acknowledged as source references and starting materials; the records do not establish that the team authored the original designs or that any specific open-source license applies.

The documented adaptation was to integrate the room functions into a coordinated room controller, create a separate tank controller, add a pump and protected driver circuit plus AUTO and MANUAL FILL controls, define tank-level control behavior, and develop a room-and-tank CAD presentation with a servo-driven sliding entry, visible tank gauge and controls, and placement for sensors and electronics [R6–R10]. The CAD notes describe layout and motion visualization, not a validated construction drawing or live electronic simulation.

The team should identify the three applications built by its four members by name and state whether any app code, UI assets, libraries, APIs, or templates were borrowed. No app names, repositories, licenses, or asset lists are established by the accessible project records. Do not claim that all project elements were created independently until the team has completed that check.

1. Scope and evidence basis

This disclosure answers what outside designs or resources informed the project and what the team changed. It is grounded in the project records located in the available Smart Home workspace and in the user-provided context that the project has three applications developed across four team members, with the submitting student responsible for Fusion 360 room/CAD work. The referenced ChatGPT conversation contains only the disclosure question and no attachments. The named Smart_Home_Participant_Brief and PS_Track2 PDFs were not available in the inspected workspace; accordingly, this report does not quote or infer requirements from them.

"Documented" below means the details appear in an available project file. It does not independently prove who created a source, which source revision was used, or that a physical circuit or CAD motion was tested. "Confirm" marks a fact the team must verify before making a definitive submission claim.

Evidence group	What the available records establish	What remains unverified
Tinkercad references	Four named feature-circuit links are listed as original source links; project notes call them source references.	Account/creator identity, revision date, reuse terms/license, and whether each was copied, imported, or used only as guidance.
Tank board	A supplied board export is described as containing Uno, HC-SR04, LCD, buzzer, and LEDs, but no pump driver or control switches.	Original author, exact export/source link, date, and license.
CAD	Room/tank mesh and Fusion import/build notes exist; they specify dimensions, components, sliding entry, visible tank, and motion setup.	Native Fusion design history and authorship; any mesh/model assets used as bases; dimensions and final assembly verification.

Electronics and report	Two Arduino sketches, integration maps, build notes, and schematic PDFs are present in prior project outputs.	Third-party code/library use, schematic symbol/footprint sources, actual physical testing, and whether PDFs are original drawings or adapted graphics.
Three applications	User context states three apps across four team members.	App names, platforms, individual ownership, code and asset provenance, dependencies, and licenses.

2. Source register

The following register lists the concrete sources named in the project records. The four Tinkercad links are useful for attribution, but the links alone do not prove the identity of the account owner, date of access, reuse license, or exact state of the design when consulted. Verify the page attribution and terms as they appeared when the team used each circuit.

ID	Recorded source	Project role	License / attribution status
R1	Temperature-based fan circuit — https://www.tinkercad.com/things/dyl50oa5J0V-temperature-based-fan	Room temperature / fan feature reference.	Tinkercad source link is recorded; author, revision, and reuse terms need confirmation.
R2	Light sensor circuit — https://www.tinkercad.com/things/1M0kzhKXnJv-light-sensor	Room light-sensing feature reference.	Tinkercad source link is recorded; author, revision, and reuse terms need confirmation.
R3	Automatic door circuit — https://www.tinkercad.com/things/l2k9f3c67E7-automatic-door	Entry sensing / servo feature reference.	Tinkercad source link is recorded; author, revision, and reuse terms need confirmation.
R4	PIR motion sensor circuit — https://www.tinkercad.com/things/g8JFRvc8otv-pir-motion-sensor	Motion alert / occupancy feature reference.	Tinkercad source link is recorded; author, revision, and reuse terms need confirmation.
R5	Supplied “Copy of Water Level Motor.brd” board export (recorded in project notes).	Starting point for the tank level subsystem; board inventory lists Uno, sonar, LCD, buzzer and LEDs.	Exact origin, author, link, date, and license are absent; identify the original exporter/provider.
R6	Five circuit reference PDFs: Ultrasonic Servo; Light Sensor Lamp; PIR Motion Buzzer; Temperature Fan LCD; Ultrasonic LCD Alert.	Circuit reference/subsystem documentation stored in project outputs.	Treat as project documentation unless a drawing, symbol, or image was copied from elsewhere; trace any such components.
R7	Fusion 360 room/tank model files, including SmartHomeRoom.f3d.f3z and derived mesh/export files.	CAD room, entry and tank layout; model/build notes enumerate project-specific features.	Native design history/source assets are not recorded in the disclosure materials; confirm whether any imported base model or third-party component geometry was used.
R8	Two project Arduino sketches and their integration/build notes.	Room control and tank control behavior / pin assignments.	Check source comments, library includes, copied snippets, and authorship before asserting original code.
R9	Three team-developed applications (names to insert).	Application layer within the larger project, per team context.	No app repositories, assets, libraries or license inventory were available; complete app-by-app register below.

Application source register to complete

Application	Team member(s) / responsibility	Third-party material to record	Changes made and license
[App 1 name]	[Member(s), role]	[Frameworks, packages, APIs, UI/assets, templates, snippets]	[Specific project changes; license/notice/link]
[App 2 name]	[Member(s), role]	[Frameworks, packages, APIs, UI/assets, templates, snippets]	[Specific project changes; license/notice/link]
[App 3 name]	[Member(s), role]	[Frameworks, packages, APIs, UI/assets, templates, snippets]	[Specific project changes; license/notice/link]

3. Specific modifications documented

3.1 Room automation and circuit integration

Four separate Tinkercad feature circuits are identified as the source references. The project records then describe their room functions together under a room Arduino Uno: an ultrasonic sensor and servo-driven sliding entry; light control from an LDR; a PIR-triggered motion alert using a buzzer and LED; and temperature-based fan control with a TMP36, MOSFET and DC motor. The room LCD displays system status. The pin map assigns each feature to a defined set of Uno pins and describes the component connections [R1–R4, R8].

The integration is a documented modification in system architecture and interconnection: independent demonstrations are treated as features of one room subsystem with an explicit pin map, shared controller, status display, and defined behavior. The documentation specifies opening the door within 25 cm and returning it after 3 seconds without another near reading; turning on the light below a calibrated darkness threshold; giving one buzzer cue when motion begins while the indicator remains lit during detection; and switching the fan on at 28 °C and off at 26 °C. These are project-record values, not externally verified performance results.

The physical concept was also changed to a sideways sliding panel on rails with a slim pull handle in place of a swing-door arm/knob. The documented CAD servo mechanism is a small demonstration arrangement; it is not evidence of a full-size safe door actuator.

3.2 Tank monitoring and automatic/manual fill

The supplied tank board is explicitly described as a level-monitor circuit, not a complete pump-control system. Project notes say it has an Uno, HC-SR04, LCD, buzzer and indicator LEDs but lacks a pump, pump driver, AUTO switch and MANUAL FILL control. The project adds a pump output driven through a logic-level N-channel MOSFET, gate resistor, gate pulldown and flyback diode; a separate low-voltage pump supply with common ground; AUTO and MANUAL FILL controls; percentage calculation; and tank-specific pump logic [R5, R8].

The documented tank behavior calculates water percentage from tank depth and the sensor-to-water air gap. In AUTO, filling starts at 25% or lower and stops at 90% or higher. In manual mode, a high-level cutoff stops filling at 95%. A missing/invalid sonar echo turns the pump off and reports a sensor error. The integration notes retain a separate Uno for the tank to avoid pin conflicts with the room controller and preserve the source board's sensor/LCD layout. These thresholds and safeguards are design specifications in the records; the team should confirm them against its final submitted sketch and demonstrate them in simulation or testing before describing them as validated.

3.3 Fusion 360 room and CAD work

Project notes describe a 4.5 m × 4.5 m room with four walls, a side window, open roof frame, exterior ENTRY sign, and a horizontal sliding entry on rails. A large cylindrical tank is placed beside the room. The model includes a 65% visual sight gauge, top-mounted ultrasonic sensor, percentage LCD, AUTO and MANUAL FILL controls, pump housing and fill pipe. Room/tank sensor, fan and controller components are shown as colored placement elements [R7].

The MotionReady Fusion guide further specifies 22 colored mesh parts, component creation, grounded/fixed elements, and motion joints for a sliding door, fan rotor, tank level indicator, switches/buttons, pump impeller and valve handle. This is a documented expansion from a single visual mesh toward a component assembly that can show selected mechanical motions. The guide states that the gauge marker is visual only, does not change tank volume geometry, and that the LCD does not receive live Arduino values. The model is a layout/motion illustration, not an electrical simulation, verified fabrication drawing, or proof of physical fit.

The submitter's role is recorded by the user as Fusion 360 room/CAD work. The report should retain that personal contribution statement, but the team should confirm which specific components, mesh geometry, imported parts and final assembly operations were created or modified by that member versus supplied/generated by others. No unsupported claim is made here that a named third-party CAD model was used.

3.4 Circuit documents, code and written materials

The available package contains subsystem circuit PDFs, wiring maps, assembly notes, project summaries, and separate room and tank Arduino sketches. The records support describing these as project deliverables and integration documentation. They do not, by themselves, establish whether a particular schematic symbol, library footprint, code snippet, code library, written explanation, or diagram was authored by a team member or reused. The final disclosure should name every external library or copied code/graphic asset actually present and state the associated license and required notice.

4. Third-party software, libraries and media

A platform or tool is not automatically an open-source design source. Fusion 360, Tinkercad, Arduino IDE, component datasheets, open-source libraries, application frameworks, and physical components have different roles and terms. Mention software used as tools in the methods/acknowledgments if appropriate, but do not label the resulting design "open source" unless the team deliberately published it under an identified license.

- Libraries: inspect each sketch for #include statements and any copied helper code; record library name, version, author/maintainer, source URL, license, and any modifications. Built-in platform functions should not be described as a borrowed third-party design without a reason.
- Application dependencies: export package manifests / dependency lists for all three apps and record direct dependencies, versions, licenses, and any assets or example code adapted. Also identify hosted APIs/services and their role, even when they are not open-source.
- Images, icons, fonts, audio and interface assets: record the creator/source URL, license, retrieval date, and edits. If obtained from a stock or asset site, use the applicable asset license and retain proof of the terms.
- CAD components and electronics: distinguish original geometry/drawings from manufacturer models, downloaded CAD parts, vendor datasheet graphics, and footprints/symbols. Record the specific part/model library and applicable terms.
- Text and AI-assisted work: if project writing, code, CAD descriptions, or images were generated or materially assisted by tools, follow the course/institution disclosure policy. Do not attribute generated or collaborative work to an individual as sole work unless that contribution is verified.

5. Original contribution and attribution statement

Subject to the team checks below, the following statement is suitable for the project report:

> *"Our project used four publicly linked Tinkercad circuits as references or starting points for the room temperature fan, light sensor, automatic entry, and PIR motion features. We also used a supplied water-level monitor board as the starting point for tank sensing and display. We integrated the room features under a dedicated room controller and developed a separate tank controller. The documented tank changes add a pump driver and protection components, AUTO and MANUAL FILL controls, percentage calculation, and automatic/manual cutoff behavior. Our room/tank CAD work presents a 4.5 m × 4.5 m smart-home room with a rail-mounted sliding entry and an adjacent instrumented water tank; the assembly notes add visual motion joints for selected parts. The model is a visual layout representation and does not provide live circuit simulation or prove fabrication readiness. We have recorded the source links available to us below; the original authorship, revision, and applicable reuse terms for the referenced designs, and the provenance of any libraries or app assets, are being verified and will be credited according to the applicable terms."*

After verification, replace “publicly linked” with “publicly accessible” only if confirmed, and replace “being verified” with a completed, accurate attribution statement. Do not describe a source as open-source merely because it can be viewed or copied in a platform.

6. Validation and responsible-use boundaries

The project records include explicit behavioral requirements and wiring notes, including sensor-failure pump shutdown, pump supply separation, common grounding, flyback protection, and keeping wet components away from exposed electronics. They also state that the CAD presentation is illustrative and that the 65% gauge and LCD depiction are not live. These are useful design disclosures, but the records do not include test logs or proof that every requirement was verified in a physical build.

Area	What records specify	Evidence to attach if claiming validation
Room behavior	25 cm door trigger; 3 s return; light threshold; motion alert; 28/26 °C fan hysteresis.	Tinkercad run captures or test table showing trigger values, response and repeatability.
Tank behavior	25% AUTO start; 90% AUTO stop; 95% manual cutoff; invalid echo stops pump.	Simulation/test cases for low, target, high, manual, and sensor error conditions.
Electrical protection	MOSFET/relay driver, flyback diode, separate pump supply, common ground, current limiting, water separation.	Final schematic and bill of materials; confirm component ratings and safe enclosure for any physical setup.
CAD assembly	Room/tank scale, component placement, joints and visual sample gauge.	Final Fusion screenshots/native file; dimensions and clearance review; label model as illustrative if not fabrication-validated.
Applications	Three apps stated by the team.	App names, builds/screenshots, test status, third-party dependency and asset register.

7. Completion checklist for the team

- Attach or cite Smart_Home_Participant_Brief.pdf and PS_Track2.pdf; map the final disclosure to their exact required project features and attribution rules.
- Open each R1–R4 Tinkercad link and record creator/display name, access date, available license/reuse terms, and the specific circuit elements carried forward.
- Identify the exact origin and license/terms of the supplied water-level board; retain the source file name and a stable link or provider record.
- Compare the submitted room/tank sketches to any source code. List libraries, versions, copied snippets, authors, licenses, modifications, and required notices.
- Identify all three applications and each of the four members’ responsibilities. Complete the app-specific table and inventory dependencies, APIs, UI assets and templates.
- Confirm CAD provenance using the Fusion design history and project files. List any imported model/part geometry and state which CAD features the submitting student created or modified.
- Check schematic PDFs for third-party graphics, symbols, footprints and component diagrams. Credit original sources where reused.
- Update the source register with exact URLs, retrieval dates, versions, licenses, license links, and modification notes. Attach notices where the relevant terms require them.
- Keep the distinction between a source being viewable, a source being reused, and a source being licensed for redistribution. If a license is absent or unclear, describe the source factually and ask the course supervisor what attribution is required before redistribution.
- Ensure any final originality or testing claim matches evidence that the team can show.

Appendix A Source register fields

Copy this row for every external design, model, code component, library, image, icon, font, sound, template, API, or dataset included in the submitted project.

Field	Record for each source
Source title and creator	[Exact title; author/account/organization; if unknown, state unknown.]
Stable source URL / identifier	[Direct URL, repository release/tag/commit, board ID, or vendor part number.]
Access date and version	[Date accessed; version, revision, commit, or timestamp used.]
License / terms	[Exact license name and version; link to license/terms; or “not stated/unclear”.]
Used in	[Circuit, CAD component, code file, app, report figure, media asset, etc.]
Material used	[Specific circuit blocks, geometry, code lines/functions, assets, text, or data.]
Modifications	[Specific changes: additions, removals, rewiring, dimensions, thresholds, UI edits, refactoring.]
Notice / attribution action	[Credit location, license text, copyright notice, source file, or permission record.]
Validation	[What was tested and evidence; do not equate reference use with successful validation.]

Appendix B Available project records consulted

These records support the project-specific descriptions in this report; they are not substitutes for the missing participant brief or problem statement. Use precise final submission filenames/locations when citing the project package.

- Original_Tinkercad_Links.md — lists four feature-circuit links (R1–R4).
- SmartHome_Project_Summary.md — room/tank overview, feature behavior, two-controller architecture, limitations and submission checklist.
- SmartHome_Integration_Map.md and SmartHome_Electronics_Summary.md — pin maps, pump additions, thresholds and wiring/safety notes.
- SmartHomeRoom_Final_README.txt and MotionReady_Fusion_Build_Guide.md — CAD features, import, joint/motion setup, visual-only limitations.
- Source_Board_Exports/SOURCE_NOTES.txt — identifies five board-layout exports and explains the water-level board’s missing pump/driver/switch functions.
- Two Arduino sketches, five circuit reference PDFs, the Fusion archive, and project README files located in the earlier Smart Home output folders.